Synopsis: AUTOSAR HLD Document Analysis Assistant

Student: Ruchithra S.

Registration No: 127003217

Institution: SASTRA Deemed University

Department: Computer Science and Engineering

1. Introduction

AUTOSAR High-Level Design (HLD) documents describe software components (SWCs), ports, interfaces, runnables, basic software (BSW) dependencies and safety requirements, usually as free-form PDF text. Reviewing them by hand is slow and error-prone. This project builds a Python and Streamlit application that helps an engineer analyse such a document, with every result traceable to a source page.

2. Problem Statement

Engineers must locate architectural elements in a PDF and manually check that they agree with each other, for example that every port has a defined interface, that periodic runnables have a timing, and that an ASIL rating is matched by E2E protection. General-purpose chatbots can answer questions about the PDF, but without enforced provenance their answers cannot be traced to a page.

3. Objectives

1. Ingest an HLD PDF and preprocess it into page-aware chunks.
2. Extract a structured architecture model with page and evidence provenance.
3. Answer engineering questions through the Gemini API, grounded in the document with page citations.
4. Apply five defined rule-based consistency checks (AR-PORT-01, AR-RUN-01, AR-RUN-02, AR-IF-01, AR-SAFE-01).
5. Provide transparent TF-IDF and Truncated SVD retrieval, and evaluate it against a labelled benchmark.

4. Methodology

The system has five stages behind a single Streamlit interface (app.py).

1. **Ingestion and preprocessing:** PyMuPDF parsing, then sliding-window chunking with header removal and statistics.
2. **Structured extraction:** SWCs, ports, interfaces, runnables, RTE events, BSW modules and safety entries, each marked Detected, Inferred or Unverified.
3. **Grounded chat:** a prompt built from document metadata, entities and page text, instructing the model to cite pages and admit insufficient evidence.
4. **Consistency checking:** deterministic rules producing PASS, WARNING, ERROR or UNVERIFIED findings with evidence and recommendations.
5. **Retrieval and evaluation:** TF-IDF with cosine similarity and a threshold, an optional Truncated SVD, and a ten-query benchmark (Precision, Recall, F1, MRR).

Tools: Python, Streamlit, PyMuPDF, NumPy, google-genai, pandas, PyYAML, python-dotenv, unittest.

5. Results Achieved

- **Testing:** all 71 automated tests passed, and the application started with HTTP 200. The Gemini client tests use offline mocks.
- **Consistency checks:** on the sample HLD, 11 findings (8 PASS, 1 WARNING, 0 ERROR, 2 UNVERIFIED).
- **Retrieval benchmark (SVD with 4 components, K = 3, τ = 0.15, 10 queries):** Precision@3 46.67%, Recall@3 100%, F1@3 62.00%, MRR 0.9500. This is identical to the TF-IDF baseline. A 2-component SVD scored lower (MRR 0.8333).

6. Scope and Limitations

The tool is an engineering document analysis assistant. It is not a certified AUTOSAR compliance tool or an ISO 26262 audit replacement. The benchmark measures retrieval against expected pages on a three-page sample. It does not measure generated-answer quality or enterprise-scale performance.

7. Conclusion

The project delivers a working, tested prototype that supports evidence traceability for extracted entities, findings and retrieved chunks, while explicitly identifying unverified information. Future work includes improved retrieval integration, a larger benchmark, and evaluation of generated answers. Future work includes retrieval-augmented chat, a larger benchmark, and evaluation of generated answers.